

Dyness Passive Balancing: Cell Imbalance, Cycle Projection and Bleeder-Hour Estimate

Quantitative estimate from two observed upper-charge balancing cycles

Author: Heiko Gerdes **Date:** 9 August 2026

Purpose

This report extends the derived high-voltage cell SOC(V) relationship to estimate each cell's charge imbalance separately for two observed balancing cycles. It projects both (1) the number of comparable charge/balance cycles and (2) the equivalent cumulative bleeder-resistor on-time required until all 16 cells are within 30 mV of the leading cell at the upper-charge reference point.

Assumptions

- Cells are sufficiently well matched that, at the same charge current and cell voltage, they can be assigned the same state of charge.
- The upper-charge voltage-to-SOC relationship is common to all cells.
- A maximum cell voltage of 3.500 V is the BMS SOC reset reference and is defined as 100.000% SOC.
- 3.600 V is the upper allowable cell voltage and approximately the point at which the charge MOSFET opens.
- Cell capacity is 280 Ah.
- A 56 ohm resistor is connected in parallel with a cell whenever its voltage exceeds the minimum cell voltage by more than 30 mV during charging.
- The instantaneous bleeder current is $I_{\text{bleed}} = V_{\text{cell}} / 56 \text{ ohm}$. Near 3.500 V this is approximately 62.5 mA.

SOC and charge-deficit model

The net electrochemical current entering cell i while its bleeder is active is:

$$I_{\text{cell},i}(t) = I_{\text{pack}}(t) - b_i(t) * V_i(t) / 56 \text{ ohm}$$

where $b_i(t)=1$ while the bleeder is active and zero otherwise. Cell charge change is obtained by integration:

$$\Delta Q_i = \text{integral } I_{\text{cell},i}(t) dt$$

Using 3.500 V as 100.000%, the inferred charge deficit relative to the leading cell is:

$$\Delta Q_i = (SOC_{\text{max}} - SOC_i) / 100 * 280 \text{ Ah}$$

From the derived SOC(V) relation, a cell at approximately 3.470 V when the leading cell is at 3.500 V corresponds to a residual deficit of about 0.162 Ah. This is used as the 30 mV convergence target.

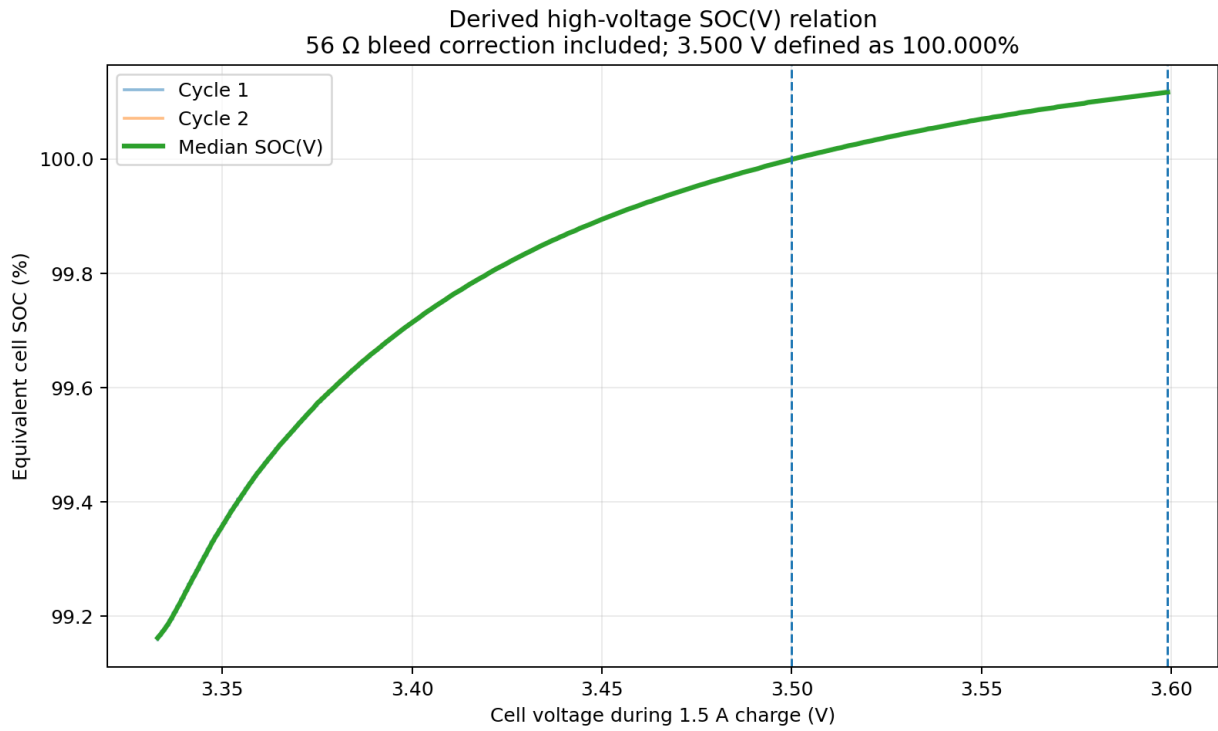


Figure 1. Derived high-voltage SOC(V) relation used for the imbalance calculation. The 3.500 V point is defined as 100.000% SOC.

Balancing progress and bleeder-hour model

Balancing is differential. The lagging cells recover relative to the leading cell because the leading cell is repeatedly shunted while the low cells are not. For cell i , the relative balancing action over one observed cycle is approximated by:

$$\Delta Q_{bal,i} = Q_{bleed,7} - Q_{bleed,i}$$

The projected number of comparable cycles required to reach the 30 mV target is:

$$N_i = (\Delta Q_i - \Delta Q_{30mV}) / (Q_{bleed,7} - Q_{bleed,i})$$

A complementary metric is the equivalent bleeder-resistor on-time. At the 3.500 V reference:

$$I_{bleed,3.5V} = 3.500 \text{ V} / 56 \text{ ohm} = 0.0625 \text{ A}$$

Therefore the idealized cumulative leading-cell bleeder time needed to remove a relative excess charge ΔQ_{excess} is:

$$t_{bleed,eq} = \Delta Q_{excess} / 0.0625 \text{ A}$$

This is an **equivalent on-time at 3.500 V**. The real resistor current varies slightly with cell voltage, so the exact elapsed clock time depends on the voltage trajectory and on how often the balancing criterion is satisfied.

Observed cycle duty

Basis	Cell-7 bleed charge	Equivalent cell-7 bleeder-on time
Cycle 1	72.9 mAh	1.17 h
Cycle 2	65.5 mAh	1.05 h

The limiting cell 11 is about 1.497 Ah behind cell 7 at the reset point. After allowing 0.162 Ah residual imbalance for the 30 mV target, approximately **1.335 Ah** of relative excess charge must still be removed. At 62.5 mA, this corresponds to **21.36 equivalent bleeder hours**.

approximately 21.4 effective leading-cell bleeder hours to the 30 mV target

Cycle 1 projection

Cycle 1 produced 72.9 mAh of cell-7 bleed charge, equivalent to about 1.17 h of continuous 56 ohm bleeder operation at 3.500 V. For cell 11, the unrounded cycle estimate is about 18.3 cycles; rounding upward gives about 19 comparable cycles. Nineteen full cycle-1-equivalent balancing exposures would correspond to about 22.2 h of accumulated cell-7 bleeder-on time, while the ideal charge-based requirement is 21.4 h.

Cycle 2 projection

Cycle 2 produced 65.5 mAh of cell-7 bleed charge, equivalent to about 1.05 h of continuous bleeder operation at 3.500 V. The unrounded cell-11 estimate is about 20.4 cycles; rounding upward gives about 21 comparable cycles. Twenty-one full cycle-2-equivalent exposures correspond to about 22.0 h of accumulated cell-7 bleeder-on time, again close to the same ~21.4 h charge-based requirement.

Per-cell imbalance, cycles and equivalent bleeder hours

Cell	C1 gap Ah	C1 cycles	C1 bleed h	C2 gap Ah	C2 cycles	C2 bleed h
1	0.708	13	8.7	0.671	15	8.1
2	0.813	13	10.4	0.773	16	9.8
3	0.250	6	1.4	0.243	11	1.3
4	0.356	10	3.1	0.339	15	2.8
5	0.325	9	2.6	0.309	14	2.4
6	1.427	18	20.2	1.404	19	19.9
7	0.000	0	0.0	0.000	0	0.0
8	1.473	18	21.0	1.451	20	20.6
9	0.684	13	8.4	0.659	16	8.0
10	0.538	12	6.0	0.509	15	5.6
11	1.497	19	21.4	1.497	21	21.4
12	1.299	16	18.2	1.258	17	17.5
13	0.538	12	6.0	0.509	16	5.6
14	1.341	17	18.9	1.278	18	17.9
15	0.913	13	12.0	0.869	15	11.3
16	0.944	14	12.5	0.929	17	12.3

The bleeder-hour columns represent the equivalent cumulative differential shunt time needed at 3.500 V to reduce that cell's relative deficit to the 0.162 Ah / 30 mV target. They are not the amount of time that the low cell's own resistor must be active. For the lowest cells, their own bleeders are generally inactive; the required correction is produced by bleeding the higher cells, especially cell 7.

The main low-SOC group remains cells 6, 8, 11, 12 and 14. Cell 11 is limiting in both cycles at about 21.4 equivalent bleeder hours.

Observed voltage progression

Cell	Cycle 1 reset	Cycle 2 reset
6	3.364 V	3.365 V
8	3.362 V	3.363 V
10	3.422 V	3.425 V
12	3.370 V	3.372 V
14	3.368 V	3.371 V

Cell 11 remained approximately 3.361 V in both reset events. This demonstrates why direct extrapolation from only the maximum voltage spread is inadequate: significant differential charge can be removed before a low cell produces a large voltage increase in the steep upper-SOC region.

Projected convergence

Basis	Cell-7 bleed/cycle	Eq. bleeder time/cycle	Limiting cell	Projected cycles
Cycle 1	72.9 mAh	1.17 h	11	~19
Cycle 2	65.5 mAh	1.05 h	11	~21

Cycle 1 includes more balancing exposure before the SOC reset than cycle 2. Cycle 2 is therefore the more conservative basis if future cycles resemble the shorter repeated charge/discharge procedure.

Best working estimate: about 20-21 comparable balancing cycles, corresponding to about 21-22 cumulative effective cell-7 bleeder hours.

The distinction between **cycle count** and **bleeder hours** is important. Cycle count depends strongly on how long the BMS permits balancing during each charge attempt. The charge-based bleeder-hour requirement is much more fundamental: approximately 1.335 Ah of differential excess charge must be removed, which at about 62.5 mA requires roughly 21.4 hours of actual shunt conduction. If future cycles provide only half as much balancing time, the required number of cycles would roughly double even though the total required bleeder-on time remains nearly unchanged.

Interpretation and limitations

- The 21.4 h value is referenced to 3.500 V. Since the actual shunt current is $V/56\text{ ohm}$ and cell voltage is somewhat lower during part of balancing, real conduction time will be slightly longer for the same removed charge.
- The convergence will not remain perfectly linear. As cells converge, the set of cells satisfying the 30 mV balancing criterion changes, so individual resistor duty cycles will change.
- The calculation assumes the derived SOC(V) relationship is transferable between matched cells at the same charging current.
- The projected endpoint is the condition that every cell lies within 30 mV at the upper-charge reference point, not perfect equality of SOC.

Conclusion

The two observed cycles independently indicate a worst-case imbalance of approximately 1.5 Ah at the BMS 100% reset. Reaching the 30 mV target permits about 0.162 Ah residual imbalance, leaving approximately 1.335 Ah to be removed differentially. With a 56 ohm bleeder near 3.5 V, the leading-cell shunt current is approximately 62.5 mA, giving a physically useful requirement of about **21.4 cumulative bleeder-resistor hours**. Under the observed duty cycles this maps to approximately **19 cycle-1-equivalent cycles** or **21 cycle-2-equivalent cycles**. A practical expectation remains roughly 18-25 cycles, but the total effective shunt-conduction requirement is around 21-22 hours.